

Amazon Sales Analysis

SQL Data Analyst Portfolio Project

A practical SQL case study using an Amazon India sales dataset.

1. Business Problem

An e-commerce business needs to understand where revenue is coming from, which products and categories perform best, how fulfilment affects sales, where customers are concentrated, and how cancellations affect operations. This project answers those questions using SQL.

2. Dataset

Metric	Value
Rows	128,975
Unique orders	120,378
Unique SKUs	7,195
Date range	31-Mar-2022 to 29-Jun-2022
Successful revenue	■6.90 crore
Successful orders	99,392
Successful units	106,895
Cancellation rate	14.21%

3. Analysis Approach

The analysis is separated into data-quality checks and business analysis. Core sales KPIs use Shipped and Shipped - Delivered to Buyer statuses. Cancelled and returned statuses are analyzed separately. This prevents cancelled/returned rows from being treated as successful sales.

4. Key Business Findings

- Set is the highest-revenue category in the supplied dataset.
- MAHARASHTRA is the highest-revenue state.
- Amazon fulfilment contributes the largest successful revenue.
- Monthly analysis shows the strongest revenue period and allows month-over-month performance to be measured.
- The cancellation rate is material enough to be treated as a separate operational KPI.
- Category, SKU, state, city and fulfilment analysis can be used to identify where management attention should be focused.

5. SQL Techniques Demonstrated

- Aggregations and GROUP BY
- COUNT(DISTINCT) for order-level KPIs
- CASE WHEN for B2B/B2C segmentation
- CTEs for readable multi-step analysis
- LAG() for month-over-month growth
- DENSE_RANK() for top products within categories
- ROW_NUMBER() for top cities within states
- Window SUM() for running totals and revenue share
- HAVING for volume-based product filtering

6. Portfolio Business Questions

1. What is successful revenue, order count, unit count and average order value?
2. How does revenue change month by month?
3. What is month-over-month revenue growth?
4. Which categories contribute the most revenue?
5. Which SKUs are the top performers?
6. Which states and cities generate the most revenue?
7. How do Amazon and Merchant fulfilment compare?
8. What is the cancellation rate by category?
9. Which products rank in the top three within each category?
10. Which SKUs have high volume but comparatively low revenue per order?

7. Interview-ready Explanation

I worked on an Amazon sales analysis project using MySQL. I started by checking row counts, unique orders, missing values and status distribution. I then defined successful sales using the relevant shipped statuses. After that I built KPI, monthly, category, product, geography, fulfilment and cancellation analyses. For advanced analysis I used CTEs and window functions such as LAG, ROW_NUMBER and DENSE_RANK. The purpose was to convert raw transaction data into business-level findings.

8. Important Portfolio Note

A strong portfolio project is not just a large SQL file. The candidate should understand the business definition, data-quality checks and reasoning behind every important query. The SQL in this project is structured so that it can be studied and explained in an interview.